

✓ First Dataset (used for testing purpose)

```
import pandas as pd
import numpy as np
import random
from datetime import datetime, timedelta
```

```
NUM_ROWS = 1000
NUM_COLS = 1000

np.random.seed(42)
random.seed(42)

accounts = [f"ACC_{i:05d}" for i in range(500)]
start_date = datetime(2024, 1, 1)
```

```
data = {
    "sender": np.random.choice(accounts, NUM_ROWS),
    "receiver": np.random.choice(accounts, NUM_ROWS),
    "amount": np.round(np.random.lognormal(mean=8, sigma=1.2, size=NUM_ROWS), 2),
    "timestamp": [
        start_date + timedelta(minutes=random.randint(0, 500000))
        for _ in range(NUM_ROWS)
    ],
    "transaction_type": np.random.choice(
        ["UPI", "NEFT", "RTGS", "SWIFT"], NUM_ROWS
    ),
    "country": np.random.choice(
        ["India", "UAE", "UK", "Singapore", "Panama"], NUM_ROWS
    ),
    "is_cross_border": np.random.choice([0, 1], NUM_ROWS, p=[0.85, 0.15]),
    "risk_flag": np.random.choice([0, 1], NUM_ROWS, p=[0.9, 0.1]),
    "velocity_score": np.round(np.random.rand(NUM_ROWS) * 10, 3),
    "network_score": np.round(np.random.rand(NUM_ROWS) * 5, 3),
}
```

```
current_cols = len(data)

for i in range(current_cols, NUM_COLS):
    col_name = f"feature_{i}"
    data[col_name] = np.round(np.random.randn(NUM_ROWS), 4)
```

```
df = pd.DataFrame(data)

csv_path = "manual_aml_1000x1000.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)
df.head()
```

✅ Dataset created successfully!
Shape: (1000, 1000)

	sender	receiver	amount	timestamp	transaction_type	country	is_cross_border	risk_flag	velocity_score	network
0	ACC_00320	ACC_00305	12610.92	2024-05-07 10:00:00	UPI	UAE	0	0	9.149	
1	ACC_00338	ACC_00013	3012.56	2024-03-22 15:16:00	UPI	UAE	0	0	1.403	
2	ACC_00437	ACC_00015	3195.30	2024-08-24 17:02:00	SWIFT	Singapore	0	0	5.376	
3	ACC_00444	ACC_00437	2167.41	2024-01-25 23:35:00	RTGS	India	0	0	5.931	
4	ACC_00135	ACC_00426	6777.34	2024-10-11 10:31:00	RTGS	Singapore	0	0	5.435	

5 rows × 1000 columns

```
import pandas as pd
from google.colab import files

# Create DataFrame
df = pd.DataFrame(data)

# Save CSV
csv_path = "manual_aml_1000x1000.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)

# Trigger download to local machine
files.download(csv_path)
```

✅ Dataset created successfully!
Shape: (1000, 1000)

Second Dataset

```
NUM_ROWS = 800
NUM_COLS = 800

np.random.seed(42)
random.seed(42)

accounts = [f"ACC_{i:05d}" for i in range(500)]
start_date = datetime(2020, 1, 1)

data = {
    "sender": np.random.choice(accounts, NUM_ROWS),
    "receiver": np.random.choice(accounts, NUM_ROWS),
    "amount": np.round(np.random.lognormal(mean=8, sigma=1.2, size=NUM_ROWS), 2),
    "timestamp": [
        start_date + timedelta(minutes=random.randint(0, 500000))
        for _ in range(NUM_ROWS)
    ],
    "transaction_type": np.random.choice(
        ["IMPS", "NEFT", "RTGS", "SWIFT"], NUM_ROWS
    ),
    "country": np.random.choice(
        ["Australia", "Japan", "UK", "Singapore", "Russia"], NUM_ROWS
    ),
    "is_cross_border": np.random.choice([0, 1], NUM_ROWS, p=[0.80, 0.20]),
    "risk_flag": np.random.choice([0, 1], NUM_ROWS, p=[0.9, 0.1]),
    "velocity_score": np.round(np.random.rand(NUM_ROWS) * 10, 3),
    "network_score": np.round(np.random.rand(NUM_ROWS) * 5, 3),
}
```

```
import pandas as pd
from google.colab import files

# Create DataFrame
df = pd.DataFrame(data)

# Save CSV
csv_path = "Am1_800x800.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)

# Trigger download to local machine
files.download(csv_path)
```

✅ Dataset created successfully!
Shape: (800, 10)

Third Dataset

```
NUM_ROWS = 2000
NUM_COLS = 2000

np.random.seed(42)
random.seed(42)

accounts = [f"ACC_{i:05d}" for i in range(500)]
start_date = datetime(2015, 1, 1)

data = {
    "sender": np.random.choice(accounts, NUM_ROWS),
    "receiver": np.random.choice(accounts, NUM_ROWS),
    "amount": np.round(np.random.lognormal(mean=8, sigma=1.2, size=NUM_ROWS), 2),
    "timestamp": [
        start_date + timedelta(minutes=random.randint(0, 500000))
        for _ in range(NUM_ROWS)
    ],
    "transaction_type": np.random.choice(
        ["IMPS", "NEFT", "RTGS", "UPI"], NUM_ROWS
    ),
    "country": np.random.choice(
        ["Myanmar", "Bhutan", "Bangladesh", "Singapore", "Russia"], NUM_ROWS
    ),
    "is_cross_border": np.random.choice([0, 1], NUM_ROWS, p=[0.85, 0.15]),
    "risk_flag": np.random.choice([0, 1], NUM_ROWS, p=[0.9, 0.1]),
    "velocity_score": np.round(np.random.rand(NUM_ROWS) * 10, 3),
    "network_score": np.round(np.random.rand(NUM_ROWS) * 5, 3),
}

import pandas as pd
from google.colab import files

# Create DataFrame
df = pd.DataFrame(data)

# Save CSV
csv_path = "Am1_2000x2000.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)

# Trigger download to local machine
files.download(csv_path)
```

✅ Dataset created successfully!
Shape: (2000, 10)

Fourth Dataset

```

NUM_ROWS = 1500
NUM_COLS = 1500

np.random.seed(42)
random.seed(42)

accounts = [f"ACC_{i:05d}" for i in range(500)]
start_date = datetime(2018, 12, 15)

data = {
    "sender": np.random.choice(accounts, NUM_ROWS),
    "receiver": np.random.choice(accounts, NUM_ROWS),
    "amount": np.round(np.random.lognormal(mean=8, sigma=1.2, size=NUM_ROWS), 2),
    "timestamp": [
        start_date + timedelta(minutes=random.randint(0, 500000))
        for _ in range(NUM_ROWS)
    ],
    "transaction_type": np.random.choice(
        ["IMPS", "NEFT", "RTGS", "UPI"], NUM_ROWS
    ),
    "country": np.random.choice(
        ["Egypt", "Denmark", "Nepal", "Ukraine", "China"], NUM_ROWS
    ),
    "is_cross_border": np.random.choice([0, 1], NUM_ROWS, p=[0.85, 0.15]),
    "risk_flag": np.random.choice([0, 1], NUM_ROWS, p=[0.9, 0.1]),
    "velocity_score": np.round(np.random.rand(NUM_ROWS) * 10, 3),
    "network_score": np.round(np.random.rand(NUM_ROWS) * 5, 3),
}

import pandas as pd
from google.colab import files

# Create DataFrame
df = pd.DataFrame(data)

# Save CSV
csv_path = "Aml_1500x1500.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)

# Trigger download to local machine
files.download(csv_path)

```

✅ Dataset created successfully!
Shape: (1500, 10)

Fifth Dataset

```

NUM_ROWS = 2200
NUM_COLS = 2200

np.random.seed(42)
random.seed(42)

accounts = [f"ACC_{i:05d}" for i in range(500)]
start_date = datetime(2015, 7, 9)

data = {
    "sender": np.random.choice(accounts, NUM_ROWS),
    "receiver": np.random.choice(accounts, NUM_ROWS),
    "amount": np.round(np.random.lognormal(mean=8, sigma=1.2, size=NUM_ROWS), 2),
    "timestamp": [
        start_date + timedelta(minutes=random.randint(0, 500000))
        for _ in range(NUM_ROWS)
    ],
    "transaction_type": np.random.choice(
        ["IMPS", "NEFT", "RTGS", "UPI"], NUM_ROWS
    ),

```

```
    ),
    "country": np.random.choice(
        ["Israel", "Lebanon", "Iraq", "South Korea", "Italy"], NUM_ROWS
    ),
    "is_cross_border": np.random.choice([0, 1], NUM_ROWS, p=[0.85, 0.15]),
    "risk_flag": np.random.choice([0, 1], NUM_ROWS, p=[0.9, 0.1]),
    "velocity_score": np.round(np.random.rand(NUM_ROWS) * 10, 3),
    "network_score": np.round(np.random.rand(NUM_ROWS) * 5, 3),
}

import pandas as pd
from google.colab import files

# Create DataFrame
df = pd.DataFrame(data)

# Save CSV
csv_path = "Aml_2200x2200.csv"
df.to_csv(csv_path, index=False)

print("✅ Dataset created successfully!")
print("Shape:", df.shape)

# Trigger download to local machine
files.download(csv_path)
```